

5. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls

OUTPUT

```
kannamina_abhishek@LAPTOP-D9F9IT80:~$ nano p5_pipe.c
kannamina_abhishek@LAPTOP-D9F9IT80:~$ gcc p5_pipe.c -o p5_pipe
kannamina_abhishek@LAPTOP-D9F9IT80:~$ ./p5_pipe
Child received: Hello from Parent
kannamina_abhishek@LAPTOP-D9F9IT80:~$ |
```

CODE

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    int fd[2];
    pipe(fd);

    pid_t pid = fork();

    if (pid == 0) {
        char buffer[100];

        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));
        printf("Child received: %s\n", buffer);

        close(fd[0]);
    } else {
        char message[] = "Hello from Parent";

        close(fd[0]);

        write(fd[1], message, strlen(message) + 1);
        close(fd[1]);

        wait(NULL);
    }

    return 0;
}
```